

1 Prerequisites

This document (ab)uses index/einstein notation.

2 Definitions

Let d be the depth of the network.

Let $\mathbf{x}^{(0)}, \mathbf{x}^{(1)}, \dots, \mathbf{x}^{(d)}$ be the layers of the neural network. $\mathbf{x}^{(0)}$ is the input layer, and $\mathbf{x}^{(d)}$ the output layer.

Let f_k be the (element-wise) activation function acting on the k th layer.

Let $W^{(k)}$ be the matrix of weights at the k th layer, i.e.:

$$\mathbf{x}^{(k+1)} = f_k(\mathbf{x}^{(k)} W^{(k)})$$

Define $\mathbf{y}^{(k)} = \mathbf{x}^{(k)} W^{(k)}$ for convenience - so that $\mathbf{x}^{(k+1)} = f_k(\mathbf{y}^{(k)})$.

Let the loss function be L so that our loss is $L(\mathbf{x}^{(d)})$. Note L implicitly depends on our target output.

3 Derivation

We want to find the gradient of L w.r.t $W^{(k)}$, i.e. we want:

$$\begin{aligned} \frac{\partial L}{\partial W_{ij}^{(k)}} &= \frac{\partial L}{\partial y_{\alpha}^{(d-1)}} \frac{\partial y_{\alpha}^{(d-1)}}{\partial y_{\beta}^{(d-2)}} \dots \frac{\partial y_{\gamma}^{(k+1)}}{\partial y_j^{(k)}} \frac{\partial y_j^{(k)}}{\partial W_{ij}^{(k)}} \quad (\text{This line is not true index notation - don't sum over } j) \\ &= \frac{\partial L}{\partial y_{\alpha}^{(d-1)}} \frac{\partial y_{\alpha}^{(d-1)}}{\partial y_{\beta}^{(d-2)}} \dots \frac{\partial y_{\gamma}^{(k+1)}}{\partial y_j^{(k)}} x_i^{(k)} \end{aligned} \quad (1)$$

Note we can fix index of $\mathbf{y}^{(k)}$ to j as $W_{ij}^{(k)}$ only affects L through $y_j^{(k)}$

We have that as $y_a^{(k+1)} = f_k(y_b^{(k)}) W_{ba}^{(k+1)}$:

$$\frac{\partial y_a^{(k+1)}}{\partial y_b^{(k)}} = f'_k(y_b^{(k)}) W_{ba}^{(k+1)} \quad (\text{No sum again}) \quad (2)$$

4 Implementing Bias

To implement bias, we add a bias node with value 1 input into each layer. The maths here remains almost the same except $W^{(k)}$ now has an extra row compared to $\mathbf{y}^{(k)}$ - the last row of W can essentially be ignored for (2) as this only adds a constant term to $y_a^{(k)}$ (w.r.t $y_b^{(k-1)}$). However the $\mathbf{x}$ term in (1) does require the extra 1 as changing the bias weight component would affect $\mathbf{y}^{(k)}$.